AI Architect Master Course

Beginner to Advanced Generative AI & Agentic AI Architecture Program

Course Vision

This course is designed to transform:

- Software engineers
- Cloud architects
- Backend developers
- ML engineers
- Enterprise architects
- Technical managers

into:

AI/ML & Agentic AI Architects

who can:

- design enterprise-grade AI systems,
- build production-ready GenAI applications,
- architect Agentic AI ecosystems,
- implement scalable AI infrastructure,
- govern AI responsibly,
- and lead AI transformation initiatives.

Preferred Course Format (Recommended)

The course is designed in an:

“Architecture-First + System Design + Hands-On + Visual Learning” format.

Instead of only theory or coding, every topic is taught through:

1. Conceptual Foundations

- Why this exists
- Problem it solves
- Enterprise importance

- Architectural trade-offs

2. Architecture Diagrams

Every major topic includes:

- System architecture diagrams
- Component interaction diagrams
- Data flow diagrams
- Sequence flows
- Infrastructure views
- Production deployment views

3. Visual Infographics

Futuristic infographic-style explanations for:

- RAG
- Agentic AI
- MCP
- Context management
- Multi-agent systems
- AI memory systems
- LLMOps
- Observability
- Governance

4. Hands-On Labs

Practical implementation using:

- Python
- FastAPI
- LangChain
- LangGraph
- OpenAI SDK
- AWS Bedrock
- Vector DBs
- MCP Servers
- Docker
- Kubernetes

5. Enterprise Design Thinking

Every module includes:

- scalability considerations
- security
- governance
- observability

- cost optimization
- latency optimization
- reliability patterns
- domain-driven design
- modular architecture

6. Real Enterprise Use Cases

Examples from:

- Financial services
- Healthcare
- Retail
- IT operations
- Customer support
- Internal enterprise copilots
- AI automation systems

7. Portfolio Projects

Students build:

- RAG systems
- AI copilots
- multi-agent systems
- MCP servers
- autonomous workflows
- enterprise AI platforms

Learning Philosophy

This course teaches students to think like:

- AI Architect
- Platform Engineer
- System Designer
- AI Product Thinker
- Enterprise Solution Architect

NOT just prompt engineers.

COURSE STRUCTURE

PHASE 1 — AI FOUNDATIONS FOR ARCHITECTS

Goal:

Build deep understanding of AI systems from first principles.

Module 1 — AI Landscape & Evolution

Topics:

- AI vs ML vs DL vs GenAI vs Agentic AI
- Evolution of AI systems
- Foundation models
- Enterprise AI transformation
- AI ecosystem overview
- Open-source vs closed models

Architecture Diagrams:

- AI ecosystem map
- Evolution timeline
- AI capability stack

Hands-On:

- Use OpenAI APIs
- Run local LLMs
- Compare model outputs

Module 2 — LLM Fundamentals

Topics:

- Transformers
- Attention mechanism
- Tokens & embeddings
- Context windows
- Fine-tuning
- Hallucinations
- Inference pipelines
- Multi-modal models

Architecture Diagrams:

- Transformer architecture

- Attention flow
- Tokenization pipeline
- Embedding visualization

Hands-On:

- Embedding generation
- Prompt experiments
- Token analysis

Module 3 — Prompt Engineering & Reasoning

Topics:

- Prompt patterns
- Chain-of-thought
- Few-shot learning
- Structured outputs
- Tool calling
- Prompt routing
- Prompt optimization
- AI reasoning patterns

Architecture Diagrams:

- Prompt orchestration pipeline
- Reasoning chain visualization

Hands-On:

- Build structured prompting engine
- Function calling examples

PHASE 2 — GENERATIVE AI APPLICATION ARCHITECTURE

Module 4 — RAG Architecture Deep Dive

Topics:

- RAG fundamentals
- Chunking strategies
- Embeddings
- Hybrid retrieval
- Re-ranking
- Query rewriting
- Context compaction

• Multi-hop retrieval
• Knowledge graphs

Architecture Diagrams:

• Enterprise RAG architecture
• Hybrid retrieval flow
• Chunking strategies visualization
• Retrieval pipelines

Hands-On:

• Build production RAG system
• Pinecone/Weaviate/FAISS integration
• Reranking implementation

Portfolio Project:

• Enterprise knowledge assistant

Module 5 — Context Engineering & Memory Systems

Topics:

• Context management
• Memory architectures
• Session memory
• Long-term memory
• Episodic memory
• Semantic memory
• Context optimization
• Token management
• AI personalization

Architecture Diagrams:

• Memory architecture
• Context lifecycle
• Context routing system

Hands-On:

• Memory-enabled AI assistant
• Conversation memory implementation

Module 6 — MCP (Model Context Protocol)

Topics:

- MCP architecture
- MCP server design
- Tools/resources/prompts
- Protocol flows
- Modular MCP architecture
- Domain-driven MCP design
- Security patterns
- Tool orchestration
- Federated MCP systems

Architecture Diagrams:

- MCP ecosystem
- MCP request lifecycle
- Modular server architecture
- Tool orchestration architecture

Hands-On:

- Build MCP server
- Build custom MCP tools
- Integrate enterprise APIs

Portfolio Project:

- Enterprise MCP platform

PHASE 3 — AGENTIC AI SYSTEMS

Module 7 — Agentic AI Fundamentals

Topics:

- What makes AI agentic
- Planning systems
- Reflection loops
- Tool usage
- Multi-step reasoning
- Autonomous execution
- Agent lifecycle

Architecture Diagrams:

- Agent lifecycle
- Agent cognitive architecture

- Autonomous workflow system

Hands-On:

- Build simple autonomous agents

Module 8 — Multi-Agent Systems

Topics:

- Agent orchestration
- Coordinator agents
- Specialist agents
- Agent communication
- Shared memory
- Agent governance
- Collaboration patterns

Architecture Diagrams:

- Multi-agent orchestration
- Agent mesh architecture
- Distributed cognition systems

Hands-On:

- LangGraph multi-agent workflows
- OpenAI Agent SDK implementation

Portfolio Project:

- AI research team simulation

Module 9 — AI Workflow Orchestration

Topics:

- LangGraph
- State machines
- Human-in-the-loop
- Event-driven workflows
- Retry systems
- Workflow recovery
- Long-running agents

Architecture Diagrams:

- Workflow orchestration architecture
- Event-driven AI systems

Hands-On:

- Build production AI workflows

PHASE 4 — ENTERPRISE AI ARCHITECTURE

Module 10 — AI Platform Architecture

Topics:

- AI platform design
- Shared AI services
- API gateways
- AI gateways
- Model routing
- Multi-model orchestration
- AI infrastructure

Architecture Diagrams:

- Enterprise AI platform
- AI service mesh
- AI infrastructure blueprint

Hands-On:

- Build AI platform architecture

Module 11 — LLMOps & AI Operations

Topics:

- Prompt/version management
- Evaluation systems
- AI observability
- Monitoring
- Drift detection
- Tracing
- Experiment tracking
- Cost optimization

Architecture Diagrams:

- LLMOps architecture
- Observability pipelines
- Evaluation systems

Hands-On:

- LangSmith
- MLflow
- OpenTelemetry

Module 12 — AI Security, Governance & Responsible AI

Topics:

- AI security threats
- Prompt injection
- Data leakage
- AI governance
- Compliance
- PII handling
- Guardrails
- Responsible AI

Architecture Diagrams:

- Security architecture
- Governance control plane
- AI risk framework

Hands-On:

- Guardrail implementation
- Security testing

PHASE 5 — SCALABLE AI INFRASTRUCTURE

Module 13 — Cloud-Native AI Systems

Topics:

- Kubernetes for AI
- GPU infrastructure
- Serverless AI
- Scaling inference
- Distributed systems
- Streaming AI
- Edge AI

Architecture Diagrams:

- Scalable inference architecture

• GPU cluster architecture
• Multi-region AI deployment

Hands-On:

• Deploy AI systems on Kubernetes

Module 14 — AI Data Architecture

Topics:

• Vector databases
• Data pipelines
• Semantic indexing
• Data governance
• Real-time pipelines
• Feature stores
• Knowledge systems

Architecture Diagrams:

• AI data platform
• Semantic search architecture

Hands-On:

• Build AI data pipelines

PHASE 6 — ADVANCED AI ARCHITECTURE

Module 15 — Advanced Agentic Systems

Topics:

• Self-improving agents
• Reflection systems
• Planning algorithms
• Autonomous memory evolution
• AI operating systems
• Agent swarms
• Cognitive architectures

Architecture Diagrams:

• Autonomous cognitive systems
• Recursive agent architecture

Hands-On:

- Advanced autonomous agent platform

Module 16 — AI System Design Interviews

Topics:

- AI architecture interviews
- System design patterns
- Trade-off analysis
- Cost vs latency
- Security vs usability
- Enterprise AI design thinking

Architecture Diagrams:

- Interview system design templates

Hands-On:

- Mock architecture interviews

CAPSTONE PROJECTS

Project 1 — Enterprise RAG Platform

Features:

- Hybrid retrieval
- Re-ranking
- Query rewriting
- Context compression
- Multi-source ingestion
- Observability

Project 2 — Agentic AI Research Assistant

Features:

- Multi-agent collaboration
- Web research
- Memory systems
- Planning engine
- Autonomous execution

Project 3 — Enterprise MCP Platform

Features:

- Modular tool architecture
- DDD-inspired domains
- Secure connectors
- Governance layer
- Monitoring

Project 4 — AI Operations Platform

Features:

- LLM monitoring
- Cost tracking
- Evaluation
- Tracing
- Governance

WHAT STUDENTS WILL BUILD

By the end of the course students will have:

- Production-grade RAG systems
- AI copilots
- Agentic AI workflows
- MCP servers
- Multi-agent systems
- AI memory systems
- Enterprise AI architectures
- AI observability systems
- AI governance platforms

TOOLS & TECHNOLOGIES COVERED

AI/LLM Frameworks

- OpenAI SDK
- LangChain
- LangGraph
- LlamaIndex
- CrewAI

- AutoGen

Cloud & Infrastructure

- AWS Bedrock
- SageMaker
- Kubernetes
- Docker
- Terraform

Vector Databases

- Pinecone
- Weaviate
- FAISS
- ChromaDB

Backend & APIs

- FastAPI
- Python
- REST APIs
- gRPC

Observability

- LangSmith
- MLflow
- OpenTelemetry
- Grafana

SPECIAL ARCHITECTURE TRACKS

Optional specialization tracks:

1. Financial Services AI Architect

- AI for wealth management
- AI compliance systems
- Risk analysis agents

2. Enterprise Copilot Architect

- Internal copilots
- Knowledge assistants
- Workflow copilots

3. AI Platform Architect

- Shared AI infrastructure
- Multi-model platforms
- Enterprise AI governance

4. Agentic Automation Architect

- Autonomous workflows
- Multi-agent orchestration
- Human-in-the-loop systems

FINAL OUTCOME

Students completing this course should be able to:

- Design enterprise AI systems
- Build scalable GenAI applications
- Architect multi-agent ecosystems
- Build MCP-based AI integrations
- Design AI governance frameworks
- Lead AI transformation initiatives
- Crack AI architect interviews
- Operate as senior AI/ML architects

RECOMMENDED DELIVERY FORMAT

Best Overall Format

Hybrid Learning System

Combine:

1. Visual architecture diagrams
2. Infographic explainers
3. Deep conceptual teaching
4. Enterprise case studies
5. Hands-on coding labs
6. System design sessions
7. Capstone projects
8. Architecture review exercises
9. Interview preparation
10. Real production deployment walkthroughs

RECOMMENDED WEEKLY STRUCTURE

Each week:

- Day 1 → Concepts
- Day 2 → Architecture diagrams
- Day 3 → Hands-on coding
- Day 4 → Enterprise design patterns
- Day 5 → Project implementation
- Day 6 → Architecture review
- Day 7 → Reflection & optimization

RECOMMENDED VISUAL STYLE

Use:

- futuristic infographic visuals
- architecture blueprints
- system flow diagrams
- orchestration maps
- layered cognitive systems
- enterprise AI dashboards

This dramatically improves learning retention for complex AI architecture topics.